

BAIE-COMEAU, QC, Canada

WMO: 718290

Lat: 49.259N

Lon: 68.146W

Elev: 130

StdP: 99.78

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.7	(c) -23.4	(d) -32.6	(e) 0.2	(f) -24.8	(g) -29.9	(h) 0.2	(i) -22.6	(j) 15.2	(k) -12.6	(l) 13.4	(m) -12.8	(n) 5.9	(o) 310	(p) 0.705

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	25.1	17.4	23.4	16.6	21.7	15.8	19.2	23.0	18.0	21.3	17.0	20.0	5.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.9	20.8	16.7	12.1	19.5	15.7	11.3	18.4	55.2	23.1	51.4	21.4	48.3	20.0	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	10.3	9.1	-28.8	28.6	2.1	1.4	-30.3	29.6	-31.5	30.4	-32.7	31.2	-34.3	32.2
			-29.1	21.3	2.0	1.2	-30.6	22.2	-31.8	22.9	-32.9	23.5	-34.4	24.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-12.5	-10.9	-5.5	1.5	8.2	13.6	16.9	16.4	11.9	5.5	-0.7	-8.0
	DBStd	11.07	6.38	6.14	5.80	3.83	3.54	3.41	2.47	2.52	3.37	3.42	4.32	5.70
	HDD10.0	3144	697	584	481	255	80	6	0	0	17	145	321	558
	HDD18.3	5588	955	818	739	505	316	147	59	71	194	397	571	816
	CDD10.0	628	0	0	0	0	23	114	213	197	74	6	0	0
	CDD18.3	32	0	0	0	0	1	5	14	10	2	0	0	0
	CDH23.3	150	0	0	0	0	5	37	61	40	7	0	0	0
	CDH26.7	13	0	0	0	0	1	6	5	2	0	0	0	0
(14) Wind	WSAvg	4.5	5.3	5.4	5.1	4.8	4.0	3.8	3.5	3.6	3.9	4.5	4.8	5.1
(15) Precipitation	PrecAvg	1011	62	57	68	79	78	94	88	92	93	97	93	93
	PrecMax	1268	107	142	117	140	128	156	124	239	232	190	148	180
	PrecMin	846	14	30	34	18	33	53	22	27	23	42	29	50
	PrecStd	114	29	27	24	37	27	29	33	50	45	37	31	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	4.2	9.2	14.3	23.7	27.2	27.0	26.6	24.3	17.8	11.2	5.4
		MCWB	3.3	3.1	3.7	7.8	13.4	17.7	19.0	19.2	17.9	12.2	9.6	4.5
	2%	DB	1.7	2.1	5.3	10.6	19.4	24.0	25.0	24.4	21.2	14.7	8.4	3.3
		MCWB	0.8	0.8	2.2	5.5	11.6	15.8	17.7	17.5	16.1	11.4	6.7	2.3
	5%	DB	-0.5	-0.1	3.5	8.5	16.2	21.5	23.3	22.6	19.1	12.4	6.6	1.6
		MCWB	-1.3	-1.3	1.1	4.3	9.8	14.5	16.9	16.7	14.8	9.5	4.8	0.7
	10%	DB	-3.1	-1.9	2.0	6.7	13.8	19.4	21.6	20.9	17.2	10.5	5.0	0.0
		MCWB	-3.9	-3.0	0.0	3.2	8.8	13.5	16.1	15.6	13.6	7.9	3.6	-0.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.4	5.2	8.9	15.3	19.3	21.0	20.4	19.2	14.1	10.2	4.9
		MCDB	3.7	3.9	6.7	12.4	20.2	23.9	24.7	25.0	22.7	16.1	11.2	5.2
	2%	WB	0.9	1.0	2.8	6.4	12.8	17.3	19.4	18.7	17.1	12.0	7.2	2.6
		MCDB	1.5	1.7	4.6	9.3	17.3	21.3	23.1	22.6	19.9	14.2	7.9	3.1
	5%	WB	-1.4	-1.2	1.5	4.9	11.1	15.9	18.2	17.6	15.4	10.0	5.1	0.9
		MCDB	-0.6	-0.1	3.1	7.8	14.7	19.4	21.3	20.8	18.2	11.7	6.1	1.5
	10%	WB	-3.8	-2.9	0.1	3.6	9.6	14.6	17.1	16.7	14.1	8.3	3.7	-0.7
		MCDB	-3.1	-2.0	1.9	6.1	12.9	18.3	20.1	19.4	16.6	10.1	4.9	0.1
(35) Mean Daily Temperature Range	MDBR		8.3	8.5	8.3	7.2	7.9	8.0	7.7	7.6	7.4	5.9	5.3	6.8
	5% DB	MCDBR	8.5	7.5	8.3	9.7	12.2	11.7	10.6	10.3	9.9	8.4	6.3	6.8
		MCWBR	8.5	6.7	6.3	6.3	6.6	6.0	5.3	5.3	6.3	6.0	5.5	6.6
	5% WB	MCDBR	8.7	7.1	7.8	8.6	10.0	9.7	8.9	8.5	8.9	7.9	6.0	6.8
		MCWBR	8.8	6.7	6.6	6.3	6.6	6.0	5.5	5.2	6.5	6.5	5.9	6.7
(40) Clear-Sky Solar Irradiance	taub		0.263	0.269	0.299	0.352	0.366	0.391	0.417	0.395	0.371	0.328	0.310	0.266
	taud		2.333	2.312	2.283	2.270	2.362	2.357	2.316	2.376	2.421	2.522	2.401	2.381
	Ebn at Noon		803	888	913	893	893	869	839	842	827	814	733	749
	Edh at Noon		67	87	108	122	117	118	122	110	95	73	64	56
(44) All-Sky Solar Radiation	RadAvg		1.22	2.20	3.46	4.53	5.21	5.52	5.27	4.51	3.41	2.01	1.08	0.82
	RadStd		0.14	0.22	0.26	0.38	0.43	0.41	0.52	0.27	0.24	0.17	0.11	0.08

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	+1.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page